

Control the servo motor by using the ultrasonic sensor

Apparatus Required:

1. Arduino Uno board
2. Ultrasonic sensor
3. Servo motor
4. Jumper wires
5. Breadboard

Servo motors have three wires: power, ground, and signal. The power wire is typically red, and should be connected to the 5V pin on the Arduino or Genuino board. The ground wire is typically black or brown and should be connected to a ground pin on the board. The signal pin is typically yellow or orange and should be connected to pin 9 on the board.

Connect VCC on the ultrasonic sensor to the 5V pin on the Arduino.

Connect the Trig pin on the ultrasonic sensor to pin 12 on the Arduino.

Connect the Echo pin on the ultrasonic sensor to pin 11 on the Arduino.

Connect the GND on the ultrasonic sensor to GND on the Arduino.

I recommend! You not connect directly the servo motor to the Arduino. I suggest you use external power to the servo. SG90 Mini Servo motors can be used.

Link for video:

<https://www.youtube.com/watch?v=MU0v7y5JL5U>

Program:

```
#include <Servo.h>

// defines pins numbers
#define trigPin 12
#define echoPin 11

Servo servo;

int sound = 250;

void setup() {
  Serial.begin(9600); // Starts the serial communication
  pinMode(trigPin, OUTPUT); // Sets the trigPin as an Output
```

```

pinMode(echoPin, INPUT); // Sets the echoPin as an Input
servo.attach(9);
}
void loop() {
long duration, distance; // defines variables
// Clears the trigPin
digitalWrite(trigPin, LOW);
delayMicroseconds(2);
// Sets the trigPin on HIGH state for 10 micro seconds
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);
// Reads the echoPin, returns the sound wave travel time in microseconds
duration = pulseIn(echoPin, HIGH);
// Calculating the distance
distance = (duration/2) / 29.1;
if (distance < 10) {
Serial.println("the distance is less than 10");
servo.write(90);
delay(500);
}
if(distance>10 && distance<30){
servo.write(180);
}
else {
servo.write(0);
}
if (distance > 60 || distance <= 0){

```

```
Serial.println("The distance is more than 60");  
}  
else {  
  // Prints the distance on the Serial Monitor  
  Serial.print(distance);  
  Serial.println(" cm");  
}  
delay(500);  
}
```